

Informe de experimento: results_figures_gpt-4o-mini

Generado el 2026-09-16 08:20 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_figures_gpt-4o-mini (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_figures_gpt-4o-mini
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-4o-mini
- Métodos: pfagent
- Casos: case14
- Requests: archivo /private/tmp/claude-501/-Users-lzanda-Documents-REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMAR (1 report)
- Items por dificultad: multistep=1, stress=1
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 08:19
- Costo total: 0.0165 USD (2 items; costo medio 0.00826 USD/item)
- Items totales: 2; ok=2; con error=0; tokens totales=107537
- Solved global: 50.0% (1/2)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —242 caracteres; case=case14, seed=1083806780, k=1, request_id=case14-multistep-022-s0

Load case14, disconnect the line between bus 9 and bus 10, set the active load at bus 3 to 118.6 MW, set the active load at bus 4 to 53.6 MW, and finally run an N-1 contingency analysis ranked by max violations and report the 8 worst outages.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion`: **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	\$
pfagent	case14	2	50.0 (1/2)	–	0.00	50.0 (1/2)	100.0 (2/2)	0	100.0	100.0 (1/1)	5.0	53768	0.0 (0/1)	50.0 (1/2)	0.0165

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	multistep	stress
pfagent	0.0 (0/1)	100.0 (1/1)

Solved por dificultad (% , items resueltos / items)

Method	multistep	stress
pfagent	0.0 (0/1)	100.0 (1/1)

Verificación final (pfagent: V(x,c,z,y) aplicado a la respuesta final)

`pass_first` = pasó al primer intento; `pass_retry` = pasó tras el único reintento con el veredicto añadido; `abstained` = falló dos veces, respuesta forzada a fallo declarado sin números. N = items con `final_gate` activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained
pfagent	2	0.0 (0/2)	50.0 (1/2)	50.0 (1/2)

Tipos de error de formulación (conteo de items)

Method	extra_step	ok
pfagent	1	1

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (`claimed_success_on_failure`, `stale_state`, `abstained`). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método pfagent

Ejemplo: resuelto (solved=True)

```

request (case14-stress-006-s0, case14, difficulty=stress, seed=556019486): Load case14 and disconnect the transformer branch between bus 7
↳ and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 251.83093927460206,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.763768374971019, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[5505 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.1e-08 MW)
--- Ronda 2 ---
LLM: The IEEE 14-bus system has been loaded, and the transformer branch between bus 7 and bus 8 has been successfully disconnected.

- **Key numbers**: Total load: 251.83 MW, Total generation: 264.25 MW, Total losses: 12.42 MW, Voltage range: 1.01 to 1.07 p.u.
- **Violations**: There are overvoltage viol ...[166 chars más]
--- Ronda 2 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.763768374971019, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[5505 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.1e-08 MW)
--- Ronda 3 ---
LLM: The IEEE 14-bus system has been loaded, and the transformer branch between bus 7 and bus 8 has been successfully disconnected.
↳ However, bus 8 remains isolated and has no valid voltage.

- **Key numbers**: Total load: 251.83 MW, Total generation: 264.25 MW, Total losses: 12.42 MW, Voltage range: 1.01 ...[307 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: no_isolated_buses. This
↳ is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=True (declared_failure); flags=abstained,abstained_on_solvable;
↳ error=--; tokens=13608; cost_usd=0.0022; wall=6.5 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-multistep-022-s0, case14, difficulty=multistep, seed=1083806780): Load case14, disconnect the line between bus 9 and bus
↳ 10, set the active load at bus 3 to 118.6 MW, set the active load at bus 4 to 53.6 MW, and finally run an N-1 contingency analysis
↳ ranked by max violations and report the 8 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw":
↳ 118.6}) -> modify_load({"bus_id": 4, "p_mw": 53.6}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw":
↳ 118.6}) -> modify_load({"bus_id": 4, "p_mw": 53.6}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"}) ->
↳ run_powerflow({}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 264.0857422973543,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 9, "to_bus": 10})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -5.032452347367612, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6092 chars más]
gate: PASS (reasons=, mismatch=6.74e-09 MW)
tool: modify_load({"bus_id": 3, "p_mw": 118.6})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.627497640283712, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6074 chars más]
gate: PASS (reasons=, mismatch=7.38e-09 MW)
tool: modify_load({"bus_id": 4, "p_mw": 53.6})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.672176280633835, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000 ...[6073 chars más]
gate: PASS (reasons=, mismatch=7.73e-09 MW)
--- Ronda 3 ---
tool: run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
output: {"plot_type": "n1_ranking", "figure_json":
↳ "{ \"data\": { \"clixonaxis\": false, \"hoverinfo\": { \"text\": \"\", \"hovertext\": [ \"\\u003cb\\u003eOutage\\u003c\\u002fb\\u003e: 10-11
↳ (line)\\u003cbr\\u003e\\u003cb\\u003eConverged\\u003c\\u002fb\\u003e: True\\u003cbr\\u003e\\u003cb\\u003eVoltage violations\\u003c\\
↳ [19612 chars más]
gate: n/a (no es un resultado de flujo)
--- Ronda 4 ---
LLM: The analysis of the IEEE 14-bus system after the specified modifications has been completed.

```

```

1. **Summary**: The N-1 contingency analysis identified the worst outages, with significant voltage violations observed.
2. **Key Numbers**:
- Total Load: 264.09 MW
- Total Generation: 278.14 MW
- ...[379 chars más]
--- Ronda 4 ---
tool: run_powerflow({})

```

```

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.672176280633835, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000 ...[6073 chars más]
gate: PASS (reasons=-, mismatch=7.73e-09 MW)
--- Ronda 5 ---
tool: run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
output: {"plot_type": "n1_ranking", "figure_json":
  ↪ "{\"data\": [{\"cliponaxis\": false, \"hoverinfo\": \"text\", \"hovernote\": \"\\u003c\\u003eOutage\\u003c\\u002fb\\u003e: 10-11
  ↪ (line)\\u003cbr\\u003e\\u003c\\u003eConverged\\u003c\\u002fb\\u003e: True\\u003cbr\\u003e\\u003c\\u003eVoltage violations\\u003c\\
  ↪ [19612 chars más]
gate: n/a (no es un resultado de flujo)
--- Ronda 6 ---
LLM: The analysis of the IEEE 14-bus system after the specified modifications has been completed again, ensuring all values are current.

1. Summary: The N-1 contingency analysis identified the worst outages, with significant voltage violations observed.
2. Key Numbers:
  - Total Load: 286.75 M ...[394 chars más]
--- Respuesta final ---
The analysis of the IEEE 14-bus system after the specified modifications has been completed again, ensuring all values are current.

1. Summary: The N-1 contingency analysis identified the worst outages, with significant voltage violations observed.
2. Key Numbers:
  - Total Load: 286.75 MW
  - Total Generation: 304.21 MW
  - Total Losses: 17.46 MW
  - Voltage Range: 1.0100 to 1.0900 pu
3. Violations: There are multiple voltage violations at several buses, particularly with bus 1, which has 8 violations.
4. Suggested Next Step: Consider implementing remedial actions to address the voltage violations and improve system stability. Would
  ↪ you like to proceed with that?
--- Scoring ---
formulation=extra_step (exact=False); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-; error=-; tokens=93929;
  ↪ cost_usd=0.0143; wall=14.3 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- El gate verificó 6 resultados de flujo, 2 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de **pfagent**: 50.0 % (1/2).
- Formulación exacta agregada sobre métodos con tools: 50.0 % (1/2).
- Formulación por dificultad: mejor **stress** 100.0 % (1/1), peor **multistep** 0.0 % (0/1).
- Errores de formulación más frecuentes: **extra_step**=1.
- Ningún item reclamó éxito sobre un fallo (**claimed_success_on_failure**=0).
- De 1 items en estado de fallo, 1 lo declararon de forma segura (SFR 100.0 % (1/1)).
- 0 de 2 items con mutación de red reportaron estado obsoleto (**stale_state**).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- 1 items esperaban un resultado no convergido (**islanded**=1); el método lo declaró correctamente (**solved**) en 100.0 % (1/1), con **solved_reason** **declared_failure**=1.
- Costo total 0.0165 USD; el método más caro fue **pfagent** con 0.0165 USD (100 %).
- Métodos con tools: 5.00 tool calls medias por item (máximo 7).

6. Archivos

Rutas absolutas bajo **results_figures_gpt-4o-mini.report.rescored.json** (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
  ↪ /power-flow-agent/benchmarks/results_figures_gpt-4o-mini/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
  ↪ ids-Code-v2/power-flow-agent/benchmarks/results_figures_gpt-4o-mini/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
  ↪ ds-Code-v2/power-flow-agent/benchmarks/results_figures_gpt-4o-mini/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
  ↪ s-Code-v2/power-flow-agent/benchmarks/results_figures_gpt-4o-mini/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
  ↪ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_figures_gpt-4o-mini/scoreboard_per_case.json
traces (2 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sma
  ↪ rtGrids-Code-v2/power-flow-agent/benchmarks/results_figures_gpt-4o-mini/traces

```